



Why Raw Beat Scaffolded

A per-angle analysis of how each module's scaffolding affected exploration, overfitting, and canonical results — webinar-AI / idea-01, run 20260527-164838

■ Executive Summary

Across 75 scaffolded agents and 10 raw agents, raw out-performed 14 of 15 scaffolded families on canonical RMSE. The scaffolding was doing its job — passing rubric, declaring contracts, structuring reports — but it was simultaneously acting as a *diversity sink* and an *in-sample evaluation trap*. Raw agents had to search; scaffolded agents had a ladder to climb. raw mean $\Delta\%$ = +26.4

- **Convergence collapse:** three of five angles produced families with std < 5%, meaning every agent in those families re-discovered the same trick. Angle E module-3 and module-4 are pinned to +10.9% $\pm$ 0 — literal flat-lines.
- **Slice narrowing:** scaffolds frequently steered agents to fit on Mach-E only or on 80 hand-picked segments. Canonical eval pools both Ford platforms across 545 segments, so those agents were graded on data they hadn't fit. Several catastrophic regressions (−67%, −94%, −60%) trace directly to this.
- **Self-evolution didn't rescue the runs that had it.** Angle A M4 and Angle D M3 already include self-patching loops, but the feedback signal targeted *rubric pass*, not RMSE. The skill grew toward defensibility, not accuracy.
- **The one family that beat raw** (Angle A M2, mean +32.7%) is the family with the highest variance and widest range (std 18.4%, max +54.9%). It scaffolded constraints without

prescribing a recipe.

■ 1. Setup — what the agents were given

The experiment ran 85 agents on the same task: improve the lateral (yaw-rate) prediction of a kinematic single-track (KS) vehicle model against measured ground truth. Ten agents got the **raw** prompt — *“The lateral predictions from our vehicle model aren’t as good as they should be. Make them better, and tell me how much each change contributed.”* — with nothing else. The remaining 75 were spread across $5 \times 3 = 15$ scaffolded variants: five *angles* (A–E), each at three *module* levels (M2, M3, M4) that progressively added scaffolding.

Each angle represents a different philosophy of how to scaffold an agent for an engineering task. Each module within an angle represents the same philosophy applied with progressively more structure. M2 is the *lightest* scaffolding within the angle; M4 is the heaviest.

The two scoring lenses

- **Rubric pass rate** — did the agent name the truth channel, acknowledge the operating contract, break results down by regime, flag honest regressions? Procedural quality.
- **Canonical $\Delta\%$ vs V0** — every agent’s favourite model re-run against the *same* 545-segment Ford pool, scored against the *same* truth, vs the *same* V0 RMSE of 0.014740 rad/s. Outcome quality. Higher = lower RMSE = better.

The two lenses dissociate. Scaffolded agents averaged 100% rubric pass; raw agents only 68%. But on canonical $\Delta\%$, raw beat 14 of 15 scaffolded families. The scaffolding optimised process; it cost outcome.

■ 2. Headline results — the table to read first

Canonical $\Delta\%$ vs V0 (positive = lower RMSE than baseline), all 16 families:

family	n	mean $\Delta\%$	median	std	range
raw	10	+26.4%	+29.3%	20.4%	-29.3 ... +49.2
angle-A / M2	5	+32.7%	+23.9%	18.4%	+14.4 ... +54.9
angle-A / M3	5	+18.7%	+19.5%	6.8%	+10.9 ... +26.8
angle-A / M4	5	+3.5%	+10.9%	24.3%	-41.1 ... +26.7
angle-B / M2	5	-6.2%	-7.9%	39.5%	-67.7 ... +36.4
angle-B / M3	5	-12.9%	-19.6%	20.2%	-37.7 ... +10.9
angle-B / M4	5	+13.7%	+10.9%	5.7%	+10.9 ... +25.1
angle-C / M2	5	-13.0%	-12.9%	2.0%	-16.1 ... -10.9
angle-C / M3	5	+18.0%	+19.1%	2.8%	+13.1 ... +21.5
angle-C / M4	5	-3.0%	+19.1%	45.4%	-93.8 ... +21.7
angle-D / M2	5	+3.1%	+10.9%	32.5%	-60.1 ... +31.0
angle-D / M3	5	+3.0%	+10.9%	17.1%	-28.5 ... +21.5
angle-D / M4	5	+1.5%	+10.9%	31.2%	-60.1 ... +22.8
angle-E / M2	5	+11.6%	+10.9%	1.4%	+10.9 ... +14.3
angle-E / M3	5	+10.9%	+10.9%	0.0%	+10.9 ... +10.9
angle-E / M4	5	+10.9%	+10.9%	0.0%	+10.9 ... +10.9

Context: +10.9% is the canonical signature of one specific recipe — KS recalibrated with canonical wheelbase plus per-segment yaw-gyro bias subtracted on straights, applied only on Mach-E (so F-150 contributes unchanged). When a family clusters around +10.9% or pins to it exactly, that family converged on this single trick.

■ 3. Angle A — Documented Ratchet

Identity. Angle A's scaffolding philosophy is “accumulating structural constraints”. The substrate documents the operating contract, the platform-truth matrix, and a numbered

catalogue of past failures ("known traps"). M2 documents the contract; M3 crystallises a methodology skill; M4 adds an automated rubric eval.

M2 — Operating contract + warnings

A long AGENTS.md (~6.3k) with the vehicle-dynamics contract: speed clamped, steering clamped, lateral residual is the only metric, Tesla has no usable truth channel, residual sign is `pred - meas`. Seven explicit traps from past runs.

Crucially, M2 stops there. No variant ladder. No tools/. The agent has to invent the method.

Result: `+32.7% ± 18.4` — the only family to beat raw. Top run reached +54.9%. The high variance is the tell: agents explored differently. Some did residual learners, some did per-platform C_α fits, some did joint K_{us} tuning.

M3 adds — crystallised skill + reference catalogue

A `SKILL.md` with a documented `lateral-fidelity-triage` procedure. A `triage.py` helper that pre-builds regime masking at fixed thresholds (`| $\delta$ | < 0.01 rad` straight, `| $d\delta/dt$ | < 0.05 rad/s` steady), RMSE computation, KS and ST models, and C_α fitting bounded to `(5e4, 5e5)` N/rad. A reference doc bounds the variant space: V0 stock KS → V1 KS recalib → V2 ST prior → V3 ST fit.

Result: `+18.7% ± 6.8`. Mean drops by 14 points, std collapses from 18 to 7. Agents stopped exploring — they ran the prescribed ladder. The ladder happens to be a reasonable one (V3 ST fit is sound) but with capped C_α bounds and Mach-E-first fitting, peak performance was bounded.

Overfitting risk introduced at M3

- **Bounded C_α search.** `(5e4, 5e5)` is reasonable, but agents who would have tried other parameterisations now don't.
- **Fixed regime thresholds.** Hardcoding 0.01 rad and 0.05 rad/s makes regime breakdowns identical across agents, which collapses the "regime-breakdown" rubric item to a copy-paste pattern.
- **V0–V3 ladder framing.** When the substrate names the rungs, agents stop adding new ones.

M4 adds — rubric-only eval sensor

A `lateral_fidelity_eval.py` “computational sensor” that runs against `REPORT.md` and checks six rubric items (truth-channel-correct, contract-acknowledged, regime-breakdown-present, methodology-consistent, attribution-coherent, honest-regression-flagged). Agent instruction: *“If the eval fails, edit the SKILL.md and add a Ratchet R<N> rule that prevents the failure recurring.”*

Result: $+3.5\% \pm 24.3$. Mean drops a further 15 points; variance jumps. The eval did its job for rubric (100% rubric pass) but the SKILL.md ratchets grew toward defensibility, not accuracy. Worst run hit -41.1% on canonical eval.

The M4 self-evolution trap. The agent was told to evolve the skill when the eval failed. The eval measured procedural quality. So the self-evolution loop trained agents to make reports more defensible, not to make models more accurate. Adding a self-evolution loop without aligning the feedback signal to the outcome metric actively hurt.

Would more aggressive self-evolution have helped? Only if the evaluation signal had measured canonical RMSE. The mechanism was right; the target was wrong.

■ 4. Angle B — Pedagogical Narrative

Identity. Angle B teaches the agent the team’s engineering culture: ISO 8855 sign conventions, the fidelity ladder (point-mass $\rightarrow$ KS $\rightarrow$ ST $\rightarrow$ nonlinear tyre), the team’s preference for interpretable physics over black-box ML. Narrative-first scaffolding.

M2 — team culture and fidelity ladder narrative

An 8k-token AGENTS.md, prose-heavy. Quoted intent: *“We tend to favour interpretable physics-based approaches over pure black-box ML.... When trade-offs arise between interpretability and accuracy, interpretability usually wins unless the accuracy gain is decisive.”* No pre-built code. The agent absorbs the culture and uses judgment.

Result: $-6.2\% \pm 39.5$, range -67.7% to $+36.4\%$. Huge variance. The cultural framing pushed several agents toward physics-first models (linear ST, understeer K_{us}) that they then fit on Mach-E-only segments and applied without recalibration to F-150 in canonical eval — producing the -67.7% blow-up.

Overfitting risk introduced at M2

- **“Interpretability over accuracy”** as a stated cultural value pushed agents away from per-segment trick fixes that actually work on the canonical pool.
- **Fidelity ladder narrative** made agents climb to ST/Ca regardless of whether their fit generalised across platforms.

M3 adds — modular skills (sim-real-runtime + vehicle-dynamics-rlog)

AGENTS.md shrinks from 8k to 1.7k. Two skills carry the load: one for runtime contract, one for vehicle-dynamics culture. Progressive disclosure (metadata first, body on demand).

Result: $-12.9\% \pm 20.2$. Mean gets worse, not better. The skill structure encouraged agents to load `vehicle-dynamics-rlog`, follow its ST-favouring guidance, and produce physically clean models that didn’t numerically beat V0. Four of five agents shipped Ca-fit ST models on Mach-E that canonically regress on F-150.

M4 adds — mandatory Research/Plan/Implement (RPI) phase discipline

Three separate artifacts: research first (no fixes proposed), plan locked (variant ladder + attribution), then implementation must follow the plan exactly. Deviations require a “Plan dissent” note.

Result: $+13.7\% \pm 5.7$. Mean jumps 26 points, variance collapses to 5.7. RPI forced agents to commit to evaluation methodology *before* seeing the numbers, killing the “tune-on-Mach-E-and-ship” failure mode but also killing exploration; four of five agents shipped V1 (+10.9%).

RPI works as a guardrail against catastrophe, not as an accelerant toward excellence. Angle B M4’s “floor” lifted from -67% to $+10.9\%$; its “ceiling” barely moved. The plan-locking discipline is a variance reducer in both directions.

Would evolving the scaffolding have helped? Yes, but specifically: evolving the *plan template* to require canonical-pool evaluation before commitment, and to require diversity in the variant ladder (don't allow all five agents to converge on V1). Today RPI lets agents replicate; it doesn't make them disagree.

■ 5. Angle C — Encoded Failure Ratchet

Identity. Angle C is the most explicit about its philosophy: *"This file is a ratchet. Every entry below is the encoded form of a real past failure."* 10 numbered constraints, each one closing a past hole.

M2 — the ratchet, no tools

AGENTS.md lists 10 explicit constraints, including the famous "use interleaved train/test split, not contiguous" rule (item 7), label per-segment vs per-platform fits (item 8), and so on. A single ablation helper `lateral_ablation.py` is provided but generic.

Result: **-13.0% ± 2.0**. Range -16.1% to -10.9%. Everyone got worse by roughly the same amount. The ratchet over-corrected: by mandating interleaved test splits on Mach-E-only data with per-platform parameters, agents fit clean in-sample numbers that canonical pooling promptly rejected. The low std is itself the signal — everyone made the *same* mistake because the ratchet steered them all into it.

How M2's ratchet became an overfit trap

- **Item 7 ("interleaved test split")** sounds like it prevents leakage. But on a 315-segment Mach-E set, interleaved 80/20 leaves both halves near-identical, so the "test" RMSE looks great. Then canonical pooling breaks the spell.
- **No instruction to score on canonical pool.** The ratchet plugged historical holes; it didn't mandate the right evaluation surface.

M3 adds — RPI + computational evals

Phase discipline (research/plan/implement) plus two automated checks: `schema_check.py` (CSV integrity) and `baseline_rmse.py` (canonical V0 per platform).

Result: $+18.0\% \pm 2.8$. Net swing of 31 points vs M2. Two changes did the work: (a) `baseline_rmse.py` exposes the canonical baseline per platform, anchoring agents to the right slice; (b) the Plan phase forces them to commit to the canonical baseline before fitting. Five of five agents shipped a per-platform variant with bias + gain, all landing in +13–21%.

M4 adds — skills library + composable recipes

A `skills/` folder with curated procedures: `baseline-residual` (canonical V0), `ablation-study` (interleaved-split-baked-in disciplined ablation).

Result: $-3.0\% \pm 45.4$. Range -93.8% to $+21.7\%$. The skill library introduced a degree of freedom: agents could now load *just* the ablation skill and skip the canonical pooling guidance from M3. One agent fitted `L_eff` on Mach-E only and recorded -94% on canonical.

Skill modularity can leak hard-won discipline. Angle C M3 forced agents through canonical pooling. Angle C M4's skills are loadable in isolation, and at least one agent loaded the ablation skill without the canonical-baseline skill. The defensive perimeter that M3 built had a door cut in it.

Would evolving the scaffolding have helped? Angle C M4 is the strongest argument *against* unconstrained scaffolding evolution. The agent was given a library; one composition produced $+21\%$, another produced -94% . Evolving the scaffolding without locking the evaluation surface invited exactly the failure mode the M2 ratchet was originally designed to prevent.

■ 6. Angle D — Expert-Authored Skill Crystallisation

Identity. Angle D delegates authorship to a "domain expert" who writes a `SKILL.md` (v0.1), runs it, sees it fail, patches it, and ships v0.5 to the agent. The agent inherits the patched version. This is the "expert hands you a tool" angle.

M2 — single expert skill v0.1

A minimal AGENTS.md (~800 words) and a `skills/lateral-fidelity-triage/` with `SKILL.md` v0.1 and `triage.py`. The variant ladder (V0 KS → V1 KS recalib → V2 ST prior → V3 ST fit) and the regime thresholds are baked into the Python helper. The agent is told: *"Read SKILL.md metadata first; load body when relevant; run the procedure."*

Result: $+3.1\% \pm 32.5$. Wild swings — -60% to +31%. v0.1 isn't mature; agents who deviated from it got hurt, agents who followed it got the +10.9% V1 floor.

M3 adds — v0.5 + regression sensor

Same skill folder but SKILL.md is now v0.5 after two rounds of off-stage self-patching. A new `sensor.py` regression gate prevents undoing prior fixes.

Result: $+3.0\% \pm 17.1$. Variance drops by half, but mean barely budes. The skill matured, but the prescribed variant ladder is still the bottleneck.

M4 adds — second composable skill

A `regime-segmentation/` skill (v0.3) alongside the mature triage skill. Designed to compose: segment the data with one skill, run the ladder with the other.

Result: $+1.5\% \pm 31.2$. Composability didn't help. Adding a second tool didn't expand the space of variants tried — it added a step that still ended in the same V1–V3 ladder.

Why Angle D plateaus across all three modules

- **The variant ladder is hardcoded in Python.** Even a fully self-patching SKILL.md can't change `triage.py`'s function signatures. The agent literally calls `run_v1(df)`, `run_v2(df)`, `run_v3(df)`; it doesn't write its own.
- **C α bounds and regime thresholds are constants in the script.** So all agents end up with the same C α , the same regime split, the same V3.
- **The regression sensor in M3 prevents *undoing* prior fixes** but doesn't encourage finding new ones. The ratchet only goes one way.

Hard-coded tools cap the ceiling. Angle D's philosophy — let an expert crystallise the recipe — works only if the expert's recipe is at the frontier. The triage.py recipe

here gets to +10.9% reliably and not much further. The skill grew in robustness, not in capability.

Would evolving the scaffolding have helped? Only if the agent had been told to evolve `triage.py` itself, not just the surrounding SKILL.md prose. The expert-crystallisation pattern needs the agent to be empowered to *extend* the recipe, not just *run* it.

■ 7. Angle E — Hand-Built Deterministic Workflow

Identity. Angle E is the opposite of agency. M2 provides five Python scripts to run in order. The substrate is explicit: *“You do not improvise the steps, you do not skip ahead, and you do not invent new variants outside the ladder.”* M3 lifts the workflow into a skill; M4 wraps the skill in RPI phases.

M2 — five steps, in order

Quoted from `AGENTS.md`:

```
step1_load_ford_segments.py ← collect Ford sim.csv
step2_segment_by_regime.py ← add regime column
step3_compute_residuals.py ← compute V0 per-regime RMSE
step4_run_st_upgrade.py ← apply Linear-ST upgrade with fit Cα; emit
V1/V2/V3 residuals
step5_write_report_skeleton.py ← produce REPORT.md skeleton
```

Every script defaults `--platform FORD_MUSTANG_MACH_E_MK1` in its example invocation. Every variant (V1 = KS recalib + per-segment yaw-gyro bias; V2 = ST prior; V3 = ST fit) is implemented in `step4_run_st_upgrade.py` exactly.

Result: `+11.6% ± 1.4`. Range +10.9% to +14.3%. Four of five agents land at exactly +10.9%, one at +14.3%. The workflow does what it says it does: ships V1, on Mach-E.

Why Angle E is the textbook overfit-by-scaffolding case

- **Platform hardcoded in script defaults.** Every agent fits and reports on Mach-E only. F-150 doesn't get touched, so canonical pooling sees F-150 unchanged from V0.

- **V1 = per-segment yaw-gyro bias is mathematically guaranteed to improve a fixed slice.** It computes the mean error on straights per segment and subtracts it. It cannot make things worse on Mach-E, and it improves it by a stable amount — exactly the +10.9% canonical signature.
- **V2 and V3 (ST with prior/fit Cα)** sometimes regressed on Mach-E (Cα fit can peg the upper bound) and so agents pick V1 as “favourite”. The workflow even has a “pegged” flag for this; everyone reads it and downgrades.

M3 adds — workflow becomes a skill

The five scripts get lifted into `yaw-divergence-triage/SKILL.md` with `trriage.py` and a sibling `regime-comparison/` skill. AGENTS.md changes tone: *“The hand-built workflow from M2 is gone; in its place sits a skills/ folder. You are a generalist agent.”*

Result: `+10.9% ± 0.0`. Zero standard deviation. Five of five agents converged on the exact same V1 number. Lifting the workflow into a skill changed the surface area but not the recipe; the recipe is still the V1 trick on Mach-E.

M4 adds — skill + RPI phase discipline

Mandatory three-phase workflow (Research, Plan, Implement) on top of the skill. *“If you find a reason mid-Phase-3 that the plan was wrong, do not silently change course.”*

Result: `+10.9% ± 0.0`. Identical to M3. RPI doesn’t change anything because the skill they’re executing already implements the V1 lock; planning around an immovable artifact doesn’t move the artifact.

Angle E is the clearest demonstration of scaffolding-as-anti-exploration. Three modules of progressive structure produced three families that converge on the same fixed point. The +10.9% is not a result; it is the V1 trick’s mathematical fingerprint.

Would evolving the scaffolding have helped? Yes — if the agent had been told to *extend* the workflow (e.g. add `step6_run_understeer_fit.py`, add F-150 to step1 default platforms, add a `step7_per_platform_recalibration`). But the substrate’s explicit instruction is *“you do not invent new variants outside the ladder.”* That is the inverse of evolution.

8. Trends across angles

Trend 1 — M2 is usually the best module in an angle

Compare the best-mean module of each angle:

angle	best module	mean $\Delta\%$
A	M2	+32.7%
B	M4	+13.7%
C	M3	+18.0%
D	M2	+3.1%
E	M2	+11.6%

Three of five angles peak at M2 — the least scaffolded variant. Adding scaffolding past the contract and warnings reduced average outcome quality in those angles. Only Angles B and C had useful M3+ additions, and only because those additions *locked the evaluation surface* (M3 added canonical baseline tooling / forced RPI planning).

Trend 2 — variance collapses faster than mean rises

Std vs raw at each module level:

- **raw:** std 20.4%
- **angle-* / M2 average:** std 18.7% (similar to raw)
- **angle-* / M3 average:** std 10.0%
- **angle-* / M4 average:** std 22.3% (skill-library composition added variance back)

Heavier scaffolding makes runs more *similar* to each other before it makes them *better*. M3 cuts std in half across the cohort, but the mean only matches raw in two of five angles.

Trend 3 — self-evolution helps when the feedback signal is canonical

Three modules invite the agent to evolve the scaffolding:

- **Angle A M4:** evolve SKILL.md against rubric eval. +3.5%

- **Angle D M3:** inherited v0.5 from prior self-patching (against expert's judgment).

+3.0%

- **Angle C M3:** RPI plan locked against *canonical* baseline. +18.0%

The one that worked is the one where the loop closes against the same metric the agent is being graded on. Self-evolution without an outcome-aligned signal trained agents to optimise the wrong thing.

Trend 4 — hardcoded tools cap the ceiling

Angles D and E both ship Python helpers that implement V1–V3 directly. Both plateau within a tight band:

- Angle D peaks at +3.1% mean, max +31%.
- Angle E peaks at +11.6% mean, max +14.3%.

Compare Angle A M2 (no hardcoded helpers, just documentation): max +54.9%. The ceiling of a scaffolded family is bounded by the ceiling of whatever recipe its tools encode.

Trend 5 — in-sample vs canonical divergence is the failure signature

Self-reported (in-sample) vs canonical $\Delta\%$ per family:

family	self-reported $\Delta\%$	canonical $\Delta\%$	gap
raw	+26.3%	+26.4%	+0.1
angle-A / M2	+22.4%	+32.7%	+10.3
angle-B / M2	+28.4%	-6.2%	-34.6
angle-C / M2	+6.3%	-13.0%	-19.3
angle-D / M2	+19.3%	+3.1%	-16.2
angle-D / M4	+23.9%	+1.5%	-22.4
angle-E / M2	+4.1%	+11.6%	+7.5

Raw agents reported and delivered the same number. Scaffolded agents systematically over-reported (Angles B, C, D) because their in-sample slice (Mach-E only, 80 segments,

custom split) was easier than the canonical pool. The honest agents at M2–M4 were honest about their *chosen* slice; the scaffold never told them which slice mattered.

■ 9. Would encouraging scaffolding evolution have helped?

The question asked: would it have helped if we'd encouraged the agent to keep building and evolving the scaffolding it started with, with instructions on how to do it?

Three of the fifteen scaffolded modules already include explicit evolution loops (Angle A M4 rubric ratchet, Angle D M3 self-patched skill, Angle C M3 RPI replan). The pattern across them is:

Self-evolution helps if and only if the feedback signal is aligned with the evaluation metric you care about. Telling an agent “evolve the SKILL.md to pass the eval” works — the agent will faithfully optimise toward whatever the eval scores. If the eval scores rubric items, you get rubric-passing reports. If the eval scores canonical RMSE on the pooled Ford set, you get lower canonical RMSE.

Specific recommendations by angle

Angle A — replace the rubric eval with a canonical RMSE eval

The SKILL.md ratchet machinery is sound. Point it at the right target: a single Python script that loads the agent's saved coefficients/function, runs them across the 545 canonical Ford segments, and reports $\Delta\%$ vs V0. Then “evolve SKILL.md when eval regresses” becomes evolutionary pressure toward accuracy.

Angle B — require RPI Phase-2 plan to lock canonical-pool evaluation

The plan template should force the agent to commit to scoring against all 545 segments, not their chosen slice. Encourage the plan to specify *at least one* variant outside the V0–V3 ladder. Locking in canonical pool kills the –67% blow-ups; mandating one off-ladder variant restores some diversity.

Angle C — tighten skills composition rules

Skills should compose with a required base set. The `ablation-study` skill should explicitly require `baseline-residual` loaded first — with a hard precondition check. This closes the door M4 cut in the M3 defensive perimeter.

Angle D — let the agent extend triage.py, not just SKILL.md

The expert-authored recipe is good but capped. Adding instruction: *“If you can lower canonical RMSE by adding a new function to triage.py, do so and document the addition in SKILL.md.”* Combined with a canonical-RMSE regression sensor (not the current sensor), this turns the static skill into an extensible one.

Angle E — invert the “no new variants” rule

Angle E’s problem is the explicit ban on new variants. Replace it with: *“Run steps 1–5 as written, then propose and run a step 6 if you see a residual structure the ladder missed.”* Make the workflow a starting point, not a terminus. Add F-150 to the default platforms in step1.

The cross-angle principle

Scaffolding evolution is a force multiplier in the direction it points. If it points at procedural defensibility, you get defensible procedures. If it points at canonical outcome, you get better outcomes. The 75 scaffolded agents largely got the former because that’s where most of their scaffolds were pointed.

■ 10. Expert discussion

Vehicle Dynamics Engineer:

The +10.9% signature is the per-segment bias trick on Mach-E only, applied to a model that has no understeer term. Of course every constrained agent finds it — it’s the only thing the V1 rung can possibly do. Adding an understeer K_{us} term, as

several raw agents did, is what unlocks +30%+ — but you have to be willing to step outside the prescribed ladder to add it.

ML Engineer:

This is overfitting to the in-sample evaluation surface, with the scaffolding providing the surface. The angle-C M2 ratchet didn't cause overfitting in the textbook sense — agents weren't memorising training points — they were optimising against a measurement protocol (interleaved Mach-E split) that diverged from the canonical one. The fix is the same as in any ML pipeline: align the dev metric to the eval metric.

Agent Systems Designer:

The interesting finding is that procedural rubric pass and outcome quality are negatively correlated across this cohort. Raw agents at 68% rubric pass, scaffolded at 100%, but raw delivered +26% canonical mean and scaffolded delivered +6% on average. If you only graded on rubric, you would conclude scaffolding works. The grading lens you pick determines which conclusion you reach.

Experiment Designer:

The cohort design did the right thing — it included both a procedural rubric and a canonical re-execution. Without the canonical eval, this whole report would have read "scaffolding works; raw is sloppy." The canonical column flipped it. Any future cohort should include both lenses by default, and report them side-by-side rather than as separate appendices.

■ 11. Conclusions

Scaffolding helped procedure and hurt outcome. Across 75 scaffolded agents and 10 raw agents on the same task, raw outperformed 14 of 15 scaffolded families on canonical RMSE. Scaffolding optimised what it was pointed at — rubric-passing reports — at the cost of what we actually wanted — lower lateral residual.

Key findings

1. **Diversity collapses fast.** Three of five angles ended up with families clustered around the +10.9% V1 fingerprint. Two families (Angle E M3 and M4) literally pinned to it with zero standard deviation.
2. **In-sample / canonical gap is the failure surface.** Scaffolded agents reported numbers from their chosen evaluation slice (Mach-E only, 80 segments, custom split). Canonical pooling exposed gaps of −34, −22, −19 percentage points. Raw agents had no gap.
3. **Hardcoded tools cap the ceiling.** Where the scaffold ships executable recipes (Angles D and E), the max canonical $\Delta\%$ is capped near the recipe's reach. Documentation-only scaffolds (Angle A M2) preserved the ceiling.
4. **Self-evolution amplifies whatever signal you point it at.** Angle A M4 evolved toward rubric pass; Angle C M3 evolved toward canonical baseline awareness. The latter produced a 31-point swing in canonical $\Delta\%$.
5. **The only family that beat raw** (Angle A M2) is the one that documented constraints without prescribing methodology. Constraints + freedom > recipes + locks.

Important caveat. This is one task and one canonical evaluation. The lateral-fidelity task may be unusually friendly to raw exploration because the search space contains discoverable physics (understeer, per-platform C α) and the V1 trick is a known local optimum that scaffolds happen to point at. On tasks with deeper search spaces, the variance-reducing effect of scaffolding may dominate; on tasks with very narrow correct answers, scaffolding may help. This finding generalises across *this* kind of task (open-ended physics modelling with multiple valid approaches).

■ 12. Future work

Immediate next steps

1. **Re-run Angle A M4 with canonical RMSE eval** — swap the rubric-checking `lateral_fidelity_eval.py` for one that runs the agent's saved model across the 545 canonical Ford segments and reports $\Delta\%$. Same ratchet machinery, outcome-aligned target.
2. **Add a "canonical-pool" sub-skill to Angle C M4** — make it a hard precondition that any ablation skill requires the canonical-pool baseline skill loaded first, and have the precondition checked at skill-load time.
3. **Modify Angle E to invert the "no new variants" rule** — replace it with "run the workflow then propose at least one step 6 outside the ladder, fit and score on the canonical Ford pool." Add F-150 to step1 default platforms.

Architecture and methodology changes

1. **Always report rubric pass and canonical $\Delta\%$ side-by-side** — this run is only diagnosable because both columns exist. Future cohorts should never produce only the procedural column.
2. **Replace prescribed variant ladders with capability primitives** — instead of "V1 = per-segment bias, V2 = ST prior, V3 = ST fit", expose "ways to characterise the residual" (bias, gain, lag, regime-specific terms) and let the agent compose. The current ladder is also a ceiling.
3. **Require diversity in scaffolded variant ladders** — if all five agents in a family produce the same RMSE, that's a signal the scaffold over-constrained. Add a "diversity violation" check to the cohort report.
4. **Bake canonical-pool evaluation into the substrate, not the scoring layer** — agents should be running the canonical pool during development, not having it applied after submission. This closes the in-sample / canonical gap by construction.



webinar-AI · idea-01 · cohort 20260527-164838 · analysis generated 2026-05-27